

Level 1: Basics

1. Print Statement

Write a Python program to print:

```
Welcome to Python
I am learning programming
```

2. Variables

Create variables to store:

- Your name
- Your age
- Your college name
- Your CGPA

Print them in a sentence.

Expected output:

```
My name is Sowdhanya, I am 22 years old, studying in XYZ College with
CGPA 8.9
```

3. Different Data Types

Create variables for:

- Student name
- Roll number
- Percentage
- Is student graduated or not

Print the value and type of each variable.

Hint:

```
type(variable)
```

4. Simple Calculator

Take two numbers and display:

Addition:
Subtraction:
Multiplication:
Division:

Example:

```
Input:
10
5

Output:
Addition = 15
Subtraction = 5
Multiplication = 50
Division = 2
```

Level 2: Input & Type Casting

5. Age Calculator

Get birth year from the user and calculate current age.

Example:

```
Input:
2004

Output:
Your age is 22
```

6. Area of Circle

Take radius as input and calculate area.

Formula:

$$\text{Area} = 3.14 * r * r$$

Example:

```
Input:
5
```

```
Output:  
Area = 78.5
```

7. Student Mark Calculator

Get marks of 5 subjects.

Find:

- Total marks
- Average mark
- Percentage

Example:

```
Input:  
90  
85  
88  
92  
95  
  
Output:  
Total = 450  
Average = 90  
Percentage = 90%
```

Level 3: Understanding Data Types

8. Swap Two Numbers

Input:

```
a = 10  
b = 20
```

Output:

Before swapping:

```
a = 10  
b = 20
```

After swapping:

```
a = 20
```

b = 10

9. Convert Data Types

Take a number as input:

"100"

Convert it into:

- Integer
- Float

Print their types.

Expected:

```
100 <class 'int'>
100.0 <class 'float'>
```

10. Bill Calculator

A shop sells:

Product name

Price

Quantity

Input:

Laptop

50000

2

Calculate:

Output:

Product: Laptop

Total Bill: 100000

Level 4: Mini Challenges

11. Personal Information System

Get:

- Name
- Age
- Height
- Weight

Display:

Name:

Age:

Height:

Weight:

BMI:

12. Temperature Converter

Convert Celsius to Fahrenheit.

Formula:

Example:

```
Input :  
25  
Output :  
77°F
```

13. Currency Converter

Input amount in INR.

Convert into USD.

Assume:

1 USD = 85 INR

Example:

```
Input :  
850  
  
Output :  
10 USD
```

Debugging Practice

Find the error:

14.

```
age = input("Enter age:")  
print(age + 5)
```

Why error occurs? Fix it.

15.

```
name = "Sowdhanya"  
age = 22
```

```
print("My name is " + name + " and my age is " + age)
```

Fix the issue.